

CAFFOX

CONTENT

Each tablet contains: Ergotamine tartrate 1 mg and caffeine 100 mg.

DESCRIPTION

A yellow round biplane scored-tablet.

INDICATIONS

Acute attacks of migraine and related types of vascular headache unresponsive to analgesics.

RECOMMENDED DOSE

CAFFOX should be given at the first symptoms of an attack.

Adults: The first time **CAFFOX** is taken, an initial dose of 2 tablets orally is recommended. If relief is not obtained within half an hour, a further tablet should be administered; this may be repeated at half-hourly intervals, but the maximum daily dose indicated below should not be exceeded. For subsequent attacks the initial dose may be increased to 3 tablets, depending on the total dose required in previous attacks. If pain persists, additional doses of 1 tablet may be taken at half-hourly intervals up to the maximum dose (see below).

The following restriction must be observed: If supplementary antimigraine medication is required, the use of any ergotamine-containing preparations, intranasal or parenteral dihydroergotamine or sumatriptan or other 5HT₁-receptor agonists must be avoided (see *Contraindications*).

Maximum dose per attack or per day: Adults: 6 mg ergotamine tartrate = 6 tablets.

Maximum weekly dose: Adults: 10 mg ergotamine tartrate = 10 tablets.

Route of administration Oral

CONTRAINDICATIONS

- Known hypersensitivity to ergot alkaloids, caffeine, or any other components of the formulations.
- Pregnancy and breast-feeding (see *Pregnancy and Lactation*).
- Impaired peripheral circulation, obliterative vascular disease, coronary heart disease, inadequately controlled hypertension, septic conditions, shock.
- Severe renal or hepatic impairment.
- Temporal arteritis.
- Hemiplegic or basilar migraine.
- Concomitant treatment with macrolide antibiotics, HIV protease or reverse transcriptase inhibitors, azole antifungals (see *Drug Interactions*).
- Concomitant treatment with vasoconstrictor agents (including ergot alkaloids, sumatriptan and other 5HT₁ receptor agonists) (see *Drug Interactions*).

PRECAUTIONS / WARNINGS

- CAFFOX** is not recommended in patients aged under 12 years or over 65 years.
- CAFFOX** is only indicated for the treatment of acute migraine attacks and not for prevention.
- Continued daily use of **CAFFOX** or its use in excess of the recommended doses must be avoided since it may cause vasospasm.
- If contrary to recommendations ergotamine-containing drugs including **CAFFOX** are used excessively over years, they may induce fibrotic changes, in particular of the pleura and the retroperitoneum. There have also been rare reports of fibrotic changes of the cardiac valves.
- Treated patients must be informed of the maximum dose that should not be exceeded and of the initial symptoms of overdose:

- Ischaemic and trophic episodes in the extremities: onset of paraesthesia (numbness, tingling), pain or vasoconstriction, even at the usual doses, must be investigated.
- Nausea and vomiting not related to the migraine.
- Symptoms of myocardial ischaemia (precordialgia).
- As soon as the symptoms of overdose appear, treatment should be discontinued and the patient should consult his or her doctor as soon as possible.
- Long-term or excessive use is contraindicated as it may lead to fibrosis (especially pleural or retroperitoneal). Rare cases of cardiac valve fibrosis have been reported.
- Ergotism including severe symptoms of peripheral blood vessel vasoconstriction, which may be potentially fatal.
- Patients suffering from mild to moderate liver failure, especially those with hepatic cholestasis, must receive suitable follow-up.
- The excessive use of antimigraine treatment may lead to the onset of chronic daily headaches, requiring the temporary withdrawal of treatment.
- Athletes' attention must be drawn to the fact that this medicinal product contains ergotamine, which is included in the list of doping substances.
- The concomitant administration of this medicinal product and dopaminergic ergot alkaloids, alpha sympathomimetics (oral and/or nasal route), indirect sympathomimetics, crizotinib, idelalisib or enoxacin is inadvisable (see *Drug Interactions*).

DRUG INTERACTIONS

Expected interactions leading to contraindication *Potent inhibitors of CYP3A4*

– The concomitant use of cytochrome P450 3A (CYP3A) inhibitors such as macrolide antibiotics (e.g. troleandomycin, erythromycin, and clarithromycin), HIV protease or reverse transcriptase inhibitors (e.g. ritonavir, indinavir, nelfinavir, delavirdine or azole antifungals (e.g. ketoconazole, itraconazole, voriconazole) and **CAFFOX** must be avoided since this can result in an elevated exposure to ergotamine and ergot toxicity (vasospasm and ischaemia of the extremities and other tissues).

Vasoconstrictors - Ergot alkaloids have also been shown to be inhibitors of CYP3A. No pharmacokinetic interactions involving other cytochrome P450 isoenzymes are known. A few cases of vasospastic reactions have been reported among patients treated concomitantly with ergotamine-containing preparations and propranolol. Concurrent use of vasoconstrictor agents including preparations containing ergot alkaloids, sumatriptan and other 5HT₁ receptor agonists, and nicotine (e.g., heavy smoking) must be avoided since this may result in enhanced vasoconstriction.

Interactions observed leading to non-recommendation of concomitant use – Any potential increase in plasma caffeine concentrations due to an interaction with one or more other medicinal products may lead to a rise in ergotamine absorption. Caffeine is significantly metabolised by CYP1A2 and medicinal products that increase or reduce enzyme activity may affect the metabolic clearance of caffeine. Fluoroquinolones, mexiletine, fluvoxamine and oral contraceptives may increase plasma exposure to caffeine. Interactions between sympathomimetics and caffeine can cause a rise in blood pressure.

Observed interactions to be taken into account

Beta-blockers - A few cases of vasospastic reactions have been reported in patients treated concomitantly with medicinal products containing ergotamine and propranolol.

Expected interactions to be taken into account

Moderate/weak CYP3A4 inhibitors - Moderate to weak CYP3A4 inhibitors such as cimetidine, clotrimazole, fluconazole, grapefruit juice, quinupristin/dalfopristin and zileuton may also increase exposure to ergotamine and caution is required if they are used concomitantly.

Serotonin reuptake inhibitors - The concurrent use of ergotamine and serotonin reuptake inhibitors (e.g., amitriptyline), including selective agents (e.g., sertraline), may cause serotonin syndrome and requires caution.

CYP3A4 inducers - CYP3A4-inducing medicinal products (e.g., nevirapine, rifampicin) can cause a reduction in the pharmacological activity of ergotamine.

Absence of interaction

No pharmacokinetic interactions involving other cytochrome P450 isoenzymes are known.

Interactions related to ergotamine

Contraindicated combinations

- Triptans (almotriptan, frovatriptan, naratriptan, rizatriptan, sumatriptan, zolmitriptan, eletriptan): risk of hypertension, coronary artery vasoconstriction. An interval of 24 hours should be left between discontinuation of the triptan and administration of the alkaloid.
- Macrolides (except spiramycin): ergotism, with the possibility of necrosis in the extremities (decrease in hepatic elimination of ergot alkaloids).
- Protease inhibitors (amprenavir, atazanavir, fosamprenavir, indinavir, lopinavir, nelfinavir, ritonavir, saquinavir, tipranavir): ergotism, with the possibility of necrosis in the extremities (inhibition of hepatic metabolism of the ergot alkaloids).
- Reverse transcriptase inhibitors (delavirdine, efavirenz): ergotism, with the possibility of necrosis in the extremities (inhibition of hepatic metabolism of the ergot alkaloids).
- Voriconazole: ergotism, with the possibility of necrosis in the extremities (inhibition of hepatic metabolism of the ergot alkaloid).
- Quinupristin-dalfopristin (in combination): ergotism, with the possibility of necrosis in the extremities (inhibition of hepatic metabolism of the ergot alkaloid).
- Stiripentol: ergotism, with the possibility of necrosis in the extremities (inhibition of hepatic metabolism of the ergot alkaloid).
- Diltiazem: ergotism, with the possibility of necrosis in the extremities (inhibition of hepatic metabolism of the ergot alkaloid).
- Phenylpropanolamine: risk of vasoconstriction and/or episodes of hypertension.
- Triclabendazole: ergotism, with the possibility of necrosis in the extremities (inhibition of hepatic metabolism of the ergot alkaloid). An interval of 24 hours should be left between the discontinuation of treatment and ergotamine, and vice versa.

Inadvisable combinations

- Dopaminergic ergot alkaloids (bromocriptine, cabergolin, pergolide, lisuride): risk of vasoconstriction and/or episodes of hypertension.
- Alpha sympathomimetics (oral and/or nasal route) (etilephrine, midodrine, naphazoline, oxymetazoline, phenylephrine, synephrine, tetraizoline, tuaminoheptane, tymazoline): risk of vasoconstriction and/or episodes of hypertension.
- Indirect sympathomimetics (except phenylpropanolamine) (ephedrine, phenylephrine, pseudoephedrine): risk of vasoconstriction and/or episodes of hypertension.

Combinations requiring precautions for use

- Beta-blockers (propranolol, oxprenolol): ergotism; some cases of arterial spasm with ischaemia in the extremities have been observed (cumulative vascular effects). Increased clinical monitoring, especially during the first week of combination therapy.

Interactions related to caffeine

Inadvisable combinations

- Enoxacin: increase in plasma caffeine concentrations, which may lead to excitation and hallucinations due to a decrease in hepatic metabolism.

Combinations to be taken into account

- Ciprofloxacin, norfloxacin: increase in plasma caffeine concentrations, which may lead to excitation and hallucinations due to a decrease in hepatic metabolism.
- Mexiletine: increase in plasma caffeine concentrations due to the inhibition of its hepatic metabolism by mexiletine.

PREGNANCY AND LACTATION

Pregnancy

Data concerning the use of ergotamine tartrate during pregnancy are limited. This product has vasoconstrictive and oxytocic properties. Consequently, **CAFFOX** should not be used during pregnancy.

Breast-feeding

Ergot derivatives are excreted into breast milk, and therefore breast-feeding is contraindicated during treatment with this product.

Ergotamine and caffeine are excreted into breast milk. Ergotamine may lead to symptoms such as vomiting, diarrhoea, weak pulse and irregular blood pressure in neonates. **CAFFOX** is therefore contraindicated in breast-feeding mothers.

SIDE EFFECTS

The most common undesirable effects are nausea and vomiting.

System Organ Class	Side Effects
Immune system disorders	<i>Rare:</i> hypersensitivity reactions (such as skin rash, facial oedema, pruritus, urticaria, dyspnoea)
Nervous system disorder	<i>Common:</i> dizziness <i>Uncommon:</i> paraesthesia in the fingers and toes, hypoesthesia, headache <i>Not known:</i> an intensifying headache with autonomic disturbances occurs within 24-48 hours of ergotamine withdrawal and may continue for 72 hours or longer. Headache is also a recognised symptom of caffeine withdrawal.
Ear and Labyrinth disorders	<i>Rare:</i> vertigo
Cardiac disorders	<i>Uncommon:</i> cyanosis <i>Rare:</i> bradycardia, tachycardia <i>Very rare:</i> myocardial ischaemia, myocardial infarction <i>Frequency not known:</i> valve fibrosis ³
Vascular disorders	<i>Uncommon:</i> peripheral vasoconstriction ¹ <i>Rare:</i> increased blood pressure <i>Very rare:</i> gangrene
Respiratory, thoracic & mediastinal disorders	<i>Frequency not known:</i> pleural fibrosis ³
Gastrointestinal disorders	<i>Common:</i> nausea and vomiting (unrelated to migraine), abdominal pain <i>Uncommon:</i> diarrhoea <i>Frequency not known:</i> retroperitoneal fibrosis ³

Musculoskeletal and connective tissue disorders	<i>Uncommon:</i> pain in the extremities, weakness in the extremities <i>Rare:</i> myalgia
Investigations	<i>Very rare:</i> absence of pulse
Injury, poisoning and procedural complications	<i>Rare:</i> ergotism ²

¹If signs of vasospasm are observed, administration of **CAFFOX** should be discontinued.

²Depending on the dose of ergotamine, intense arterial vasoconstriction may occur (ergotism). This may lead to signs and symptoms of vascular ischaemia in the extremities or in other tissues (such as renal or cerebral vasospasm).

³Long-term or excessive use may lead to fibrosis (especially pleural or retroperitoneal fibrosis). Rare cases of cardiac valve fibrosis have been reported.

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Ergotism is defined as intense arterial vasoconstriction that produces signs and symptoms of ischaemia in the extremities or other tissues (such as renal or cerebral vasospasms). Gangrene is possible if it is not treated. Most cases of ergotism are related to chronic intoxication and/or overdose and/or an interaction.

Symptoms: Abdominal pain, confusion, light-headedness, nausea, vomiting, headache, tachycardia or bradycardia associated with hypotension, respiratory disorders, dizziness, paraesthesia, ischaemia, gangrene, seizure, shock, coma, as well as symptoms of vascular ischaemia, such as numbness, tingling, pain in the extremities, cyanosis and absence of pulse. In rare cases, patients may present myocardial infarction.

Treatment: In the case of orally ingested drug, administration of activated charcoal is recommended. In the case of very recent oral intake gastric lavage may be considered. Treatment should be symptomatic. In the event of severe vasospastic reactions, intravenous administration of a peripheral vasodilator such as nitroprusside, phentolamine or dihydralazine, local application of warmth to the affected area and nursing care to prevent tissue damage are recommended. In the event of coronary constriction, appropriate treatment such as nitroglycerine should be initiated.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Patient experiencing dizziness should not drive or operate machinery.

PHARMACODYNAMICS

Ergotamine has a tonic action on vascular smooth muscle and has a special affinity for arterial monoaminergic receptors (NA and HT), especially in the external carotid network. Ergotamine may lead to vasoconstriction by stimulating alpha-adrenergic and 5-HT receptors. It has a moderate to elevated affinity for several sub-types of serotonergic receptors, but its beneficial effect on migraine is mainly linked to its action as an agonist of 5-HT1B and 5-HT1D receptors. Ergotamine aborts attacks of migraine with or without aura by its specific vasotonic action on distended extracranial arteries. Caffeine accelerates and increases the enteral absorption of ergotamine.

PHARMACOKINETICS

Ergotamine

Absorption: Ergotamine is rapidly absorbed following oral administration.

Maximum plasma concentrations are reached 1.5 to 2 hours after administration. Studies with tritium-labelled ergotamine have shown that approximately 62 % of the oral dose is absorbed via the gastrointestinal route. Maximum plasma concentrations are reached approximately 2 hours after ingestion. Any potential increase in plasma caffeine concentrations due to an interaction with one or more other medicinal products may lead to a rise in ergotamine absorption. Caffeine is significantly metabolised by CYP1A2 and

medicinal products that increase or reduce enzyme activity may affect the metabolic clearance of caffeine.

Distribution: Oral bioavailability is 62 ± 3 %. The medicinal product has a high level of tissue affinity. Protein binding of ergotamine is 98 %. The absolute bioavailability of the parent drug is approximately 2 % following oral administration and approximately 5 % following rectal administration.

Biotransformation: Ergotamine is significantly metabolised in the liver and is a substrate of the CYP3A4 enzyme system. It has been suggested that the therapeutic effects of the medicinal product are partially linked to the active metabolites.

Elimination: The parent drug and metabolites are excreted mainly in the bile. Plasma elimination takes place in two stages, with alpha and beta half-lives of 2.7 and 21 hours, respectively. Elimination in the urine is low (4.3 ± 0.4 %). The drug is primarily eliminated via the biliary and faecal routes. The addition of caffeine speeds up excretion into the blood and increases gastric absorption by at least 44 %.

Caffeine

Absorption: Following oral administration, caffeine is rapidly and almost completely absorbed via the gastrointestinal route, the maximum concentrations reached after the oral administration of 175 mg are between 5 and 10 µg/ml. Maximum plasma concentrations are reached between 15 and 120 minutes after administration.

Distribution: Plasma protein binding of caffeine is 35 %. Tissue distribution of caffeine in the body is relatively uniform, including in the cerebrospinal fluid, breast milk, saliva and sperm. The volume of distribution is approximately 0.71 l/kg. Caffeine crosses the placental barrier.

Biotransformation: Caffeine is mostly metabolised by CYP1A2 into paraxanthine. Paraxanthine is metabolised in turn into uracil and uric acid derivatives by demethylation and hydroxylation. The plasma elimination half-life is approximately 3.5 hours.

Elimination: Metabolites are predominantly excreted in the urine. Caffeine clearance is increased by smoking.

STORAGE

Store in a cool dry place, below 30 °C.

Keep out of the reach of children.

Jauhi daripada kanak-kanak.

PACKING/ PACK SIZE:

Aluminium foil strip-pack of 10 tablets per strip, boxes of 1x10 tablets.

MANUFACTURED BY:

T.O. Pharma Co. Ltd.

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Last Reviewed: 10 March 2023